Lecture 12: Single-agent dynamics

ECON 7510

Cornell University

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Parts of notes gratuitously borrowed from Tobias Salz, Nikhil Agarwal, John Asker, Nick Buchholz, Allan Collard-Wexler, Chris Conlon, Kei Kawai, Robin Lee, Ariel Pakes, Paul Scott, and Matt Shum.

Introduction

Why Dynamics?

A few observations:

- Some prices set “too low”:
  - If you joined waiting list for Green Bay Packers season tickets today, you’d likely be on the wait list for 30+ years.
  - Labor “shortage” of 2021-2022.
- Free trials are ubiquitous.
- Governments subsidize green tech companies.
- When interest rates fall, many homeowners don’t refinance, even when the interest savings are greater than the fees.

Why Dynamics?

Dynamics we might want our models to capture

- Goodwill
- Learning-by-doing
- Durability of goods
- Investment, R&D
- Firm entry/exit

Key idea: The choice an agent makes today affects the state in future periods.

Unpacking the durable goods case

- Prices of a durable good, e.g., laundry detergent, vary from week to week.
- Static model ignores the inventory problem and intertemporal substitution.
- **Question**: Which direction would our estimate of price sensitivity be biased if we estimated demand for a durable good using a static model?

Unpacking the durable goods case

- Prices of a durable good, e.g., laundry detergent, vary from week to week.
- Static model ignores the inventory problem and intertemporal substitution.
- **Question**: Which direction would our estimate of price sensitivity be biased if we estimated demand for a durable good using a static model?
  - → Leads to a larger own price elasticity with the static model, than the long-run own price elasticity with the dynamic model.
- Another bias: Leads to a lower cross price elasticity (compared to static model):
  - Response to sales is mostly coming from people going from not buying to buying the brand on sale.
  - This leads us to predict small cross price elasticities with respect to other products, and large cross price elasticity with respect to the outside good.
  - The dynamic model rationalizes this phenomenon w/ high unobserved inventories, and conditional choice probabilities (conditional on purchasing).

Review: Solving DDC models

Dynamic programming: Stochastic payoffs, discrete-state example

Value function:

$$V(x) = \max\{v_0(x), v_1(x)\}$$

where

$$\begin{aligned} v_a(x) &= u_a(x) + \delta \mathbb{E}[EV(x_{t+1}) \mid x_t = x, a_t] \\ &= u_a(x) + \delta \sum_{x'} M^a_{x,x'} EV(x') \end{aligned}$$

and

$$a^*(x, \epsilon) = 1 \Leftrightarrow v_1(x) + \epsilon_1 > v_0(x) + \epsilon_0$$

Ex-ante expected value function (before ϵ is realized):

$$\begin{aligned} EV(x) &= \mathbb{E}\left[\max\{v_0(x) + \epsilon_0, v_1(x) + \epsilon_1\}\right] \\ &= \frac{1}{\theta} \log\left[\exp\left(\theta v_0(x)\right) + \exp\left(\theta v_1(x)\right)\right] \end{aligned}$$

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Dynamic programming: Stochastic payoffs, discrete-state example

For $k = 1, 2, \ldots$ until convergence

$$
\begin{aligned}
EV_i^{k+1} &= \frac{1}{\theta} \log \left[\exp\left(\theta v_0(x)\right) + \exp\left(\theta v_1(x)\right) \right] \\
&= \frac{1}{\theta} \log \left[\sum_{a \in \{0,1\}} \exp\left(\theta \left[u_{ai} + \delta M_i^a \cdot EV^k \right] \right) \right] \text{ for each } i = 1, \ldots, N
\end{aligned}
$$

Fixed point problem in terms of EV.

Estimating Dynamic Discrete Choice Models

Dynamic Discrete Choice

Some High Level Comments

- This literature has a strong focus on computation.
- The main problem is that the estimation, as in BLP, solves a nested fixed point.
- Unlike the demand literature, the DDC literature that deals with endogeneity and unobservable confounds is much less developed.
- Requires more restrictive assumption than static discrete choice.

Outline

Road Map for Single Agent Dynamics

1. Introduction to canonical dynamic discrete choice (DDC) model.
2. Nested fixed-point (NFXP) approach: “Brute Force”. (Rust, 1987)
3. Conditional choice probability (CCP) approach 1: Exploiting observed policies. (Hotz and Miller, 1993)
4. Conditional choice probability (CCP) approach 2: Linear representation of DDC models. (Scott, 2013)
5. MPEC: Exploiting modern computational methods. (Su and Judd, 2012)
6. DDC and persistent unobserved heterogeneity. (Arcidiacono and Miller, 2011)

Basic model

Canonical Dynamic Discrete Choice

- Discrete time t, agent i, finite or infinite horizon
- s_{it}: vector of state variables
- $a_{it} \in A = \{0, 1, ..., J\}$: discrete action
- Agent's beliefs about transition process $F(s_{it+1}|s_{it}, a_{it})$.
  - Typically assume that this belief coincides with the true transition process, i.e., *rational expectations*. (Identification without this assumption would be difficult.)
  - Typically assume this is a *Markov process*.

Basic model

Data and Primitives

Data typically used:

- N individuals forming a panel (i, t).
- Observables: actions a_{it}, observable state variables x_{it}
- Unobservables (econometric error): ϵ_{it}.
  - This error is part of the state: $s_{it} = (x_{it}, \epsilon_{it})$.

We may seek to estimate:

- Structural parameters governing preferences in $u(a, s_{it})$.
- Transition probabilities.
- Discount factor β (typically difficult to identify, more on this later).

Dynamic Programming

Assumptions on unobservables

- **Additive separability**:
  - $u(a, x_{it}, \epsilon_{it}) = u(a, x_{it}) + \epsilon_{it}(a)$.
  - One unobservable for each action, so dimensionality of ϵ_{it} is J.
- **Type-I Extreme Value (logit) errors**
  - The unobservable shocks $\{\epsilon_{it}(a), a = 1, 2, ..., J\}$ are independently distributed as Type-I E.V.
- **IID**:
  - ϵ is i.i.d. distributed over actions, agents, and time.
- **Conditional independence of x_{it}**:
  - $F(s_{it+1}|s_{it}, a) = F(x_{it+1}|x_{it}, a) \cdot G_{\epsilon}$.

John Rust (1987)

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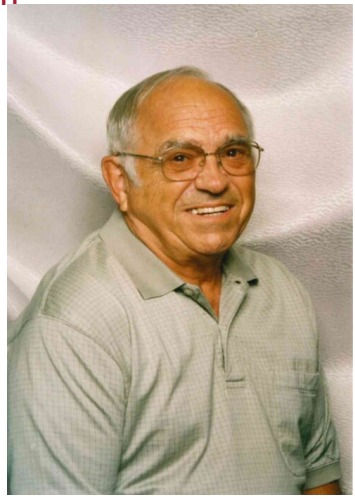
Econometrica, Vol. 55, No. 5 (September, 1987), 999-1033

OPTIMAL REPLACEMENT OF GMC BUS ENGINES: AN EMPIRICAL MODEL OF HAROLD ZURCHER

BY JOHN RUST[1]

This paper formulates a simple *regenerative optimal stopping* model of bus engine replacement to describe the behavior of Harold Zurcher, superintendent of maintenance at the Madison (Wisconsin) Metropolitan Bus Company. The null hypothesis is that

John Rust (1987) — Harold Zu(e)rcher's Problem



- He is the superintendent of maintenance at the Madison, WI Metropolitan Bus Company.
- Each week, must decide to replace engine or keep running for another week. This is an **optimal stopping** problem:
  - Fixed cost of repair (RC), but new engine has lower future maintenance costs.
  - Engine maintenance costs grows as engine gets older.
- Can show that there is a **cutoff** mileage threshold x^* above which a bus will have its engine replaced. Common type of optimal policy in many models, for example, job search.
- Acknowledgments: "I am especially grateful to Harold Zurcher for providing the data used in this study, and for his assistance in interpreting the estimation results."

John Rust (1987)

Some Comments

A good illustration of empirical IO issues and trade-offs showcased.

- Paper about one guy, who cares? The *yogurt/cereals* critique.

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- Model focuses on one particular decision and leaves out many related decisions that Harold Zurcher makes. **George E.P. Box:** *All models are wrong but some are useful…*

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- Model focuses on one particular decision and leaves out many related decisions that Harold Zurcher makes. **George E.P. Box:** *All models are wrong but some are useful…*
- However, your model should capture the crucial aspects that are material to the question. How do omitted aspects affect the conclusions drawn from the paper? It is good practice to think about all model assumptions in light of their effect on counterfactual results.

John Rust (1987)

Model

- Parameters
  - RC, θ_1: Cost parameters—replacement cost and relationship between mileage and recurring maintenance costs
  - θ_2: parameters of distribution ϵ (dist. assumed to be known)
  - θ_3: parameters that govern state transition
  - β—typically assumed.
- $a_t = 1$ if engine is replaced at time t, otherwise $= 0$. Choose sequence of actions $a \equiv \{a_1, a_2, ..., a_t, ...\}$:

$$\max_a \mathbb{E} \sum_{t=1}^{\infty} \beta^{t-1} u(x_t, \epsilon_t, a_t; \theta)$$

- Mileage x_{t+1} evolves according to $F(x_{t+1}|x_t; \theta_3)$ if $a_t = 0$, otherwise $x_{t+1} = 0$ (engine is brand new if replaced). This transition process can be estimated separately from the model.
- Utility is parameterized $u(x_t, \epsilon_t, a_t; \theta_1, RC) = -(1 - a_t)c(x_t; \theta_1) - a_t \cdot RC + \epsilon_t(a_t)$, where RC is replacement cost and c is increasing.

Rust (1987): Data & Likelihood

Conditional Independence Assumption

Allows to simplify **transition probabilities** and **value function**:

- Factor transition probabilities: $P(x_{t+1}, \epsilon_{t+1}|x_t, \epsilon_t, a_t; \theta) = P(x_{t+1}|x_t, a_t, \theta_3) \cdot G(\epsilon_{t+1}|x_{t+1}, \theta_2)$
- Do not need to treat unobserved ϵ's as state variables. Can write $EV_\theta(x_t)$ instead of $EV_\theta(x_t, \epsilon_t)$. Can express Bellman eq. in terms of EV.
- Testable: estimate alternative model where last periods decision enters utility, then perform likelihood ratio test.

Also, with **conditional independence** the **likelihood function** can be simplified to:

$$\log \ell(x_1, ..., x_T, a_1, ..., a_T|x_0, a_0; \theta) = \sum_{t=1}^{T} \log f(x_t|x_{t-1}, a_{t-1}; \theta_3) + \sum_{t=1}^{T} \log \underbrace{P(a_t|x_t; \theta_1)}_{\text{No closed form}}$$

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- Two *separable* components: the mileage transitions and the probability of replacement.

Rust (1987)

Comments on the Specification

The likelihood function is not specified directly. (It is common that we can't find a closed form expression for the maximum likelihood estimator; but in this case, we can't even find a closed for expression for the MLE *objective function*.)

- Instead, it is *derived* from the solution to the underlying optimization problem

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He discusses the importance of incorporating unobservables into the dynamic optimization:

- Model without "error term": due to cut-off rule in x_t model would be immediately rejected by the data.
- Econometric model where error term is simply added to the model solution lacks interpretation. Instead, view ϵ_t as a state variable which is unobserved to the econometrician but observed by agents (enters utility function).
  - → What is the interpretation of ϵ here?

Nested Fixed Point Algorithm

Rust (1987): Estimation

NFXP 1

Key idea: Estimation first solves an inner “loop” (the EV function) and then an outer “loop”, the objective function. Hence, nested fixed point.

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  - Goal: For given set of parameters θ_1, RC, want to compute (log) likelihood of observed choices $\ell(\{a_{it}\}; \theta_1, RC)$

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    So we need to compute choice probabilities p.
  - In order to get p, we need to solve for EV, the ex-ante expected value function EV.

Rust (1987): Estimation

NFXP 2

- Note we can specify each agent's dynamic decision problem:

$$a_t^\star = \arg\max_a \{u(x_t, \epsilon_t, a; \theta) + \beta \int EV_\theta(x_{t+1}) dF(x_{t+1} \mid x_t, a_t)\}$$

- Let $u(x, \epsilon, a; \theta) \equiv u(x, a; \theta) - \epsilon_a$ be the deterministic part of the utility minus the error term.
- Then the **choice specific value functions** are defined as:

$$v_\theta(a, x) \equiv u(x, a; \theta_1) + \beta \int EV_\theta(x') dF(x' \mid x, a)$$

- Since we assume that ϵ are i.i.d. logit:

$$P(a_t|x_t; \theta_1) = \frac{\exp\left[u(x_t, a_t; \theta_1) + \beta \int EV_\theta(x_{t+1}) dF(x_{t+1} \mid x_t, a_t)\right]}{\sum_{a'=0,1} \exp\left[u(x_t, a'; \theta_1) + \beta \int EV_\theta(x_{t+1}) dF(x_{t+1} \mid x_t, a')\right]} = \frac{\exp(v_\theta(a_t, x_t))}{\sum_{a'=0,1} \exp\left[v_\theta(a', x_t)\right]}$$

This means that if we can solve for ex ante expected value function $EV_\theta(\cdot)$, we can compute $P(a_t|x_t; \theta)$, and form the likelihood.

Rust (1987): Estimation

NFXP 3

For a given parameter value θ Rust iterates on the ex ante expected value function.

- Let x, a denote mileage and choice.
- Let y denote next period's mileage.

$$
\begin{aligned}
EV_\theta(x) &= \mathbb{E}_\epsilon \left[\max_{a=0,1} [u(x, a; \theta) + \epsilon_a + \beta \int EV_\theta(y) dF(y \mid x, a)] \right] \\
&= \log \left(\sum_{a=0,1} \exp \left[[u(x, a; \theta) + \beta \int EV_\theta(y) dF(y \mid x, a) \right] \right)
\end{aligned}
$$

- Don't need to evaluate the value function for a grid of the ϵ draws.
- Works because of **conditional independence** assumption!
- Once convergence on $EV(\cdot)$ is obtained, the choice probabilities are obtained (as on the previous slide).

Rust (1987): Estimation

Recap and Comments

Estimation Steps:

- Estimate θ_3 — transition function for mileage — can be done outside of behavioral model
- NFXP outer "loop": search over (θ_1, RC) to maximize likelihood function.
- NFXP inner "loop": for evaluating a likelihood for a given (θ_1, RC) need to find fixed point of bellman equation.

Comments:

- "Although the data clearly reject the myopic model, I was not able to precisely estimate the discount factor β." Impute value for discount factor.
- Heavy reliance on assumptions, particularly i.i.d. errors and conditional independence of state variables. Discrete states and actions, correct expectations. Will limit certain applications.
- Computationally costly: This is a full-solution method: for every evaluation of the parameter vector θ_1, the agent's dynamic programming problem is explicitly solved (value-function iteration).
- Inherits all positive aspects of maximum likelihood. Asymptotically efficient full information estimator.

Hotz and Miller (1993)

+

Hotz, Miller, Sanders and Smith (1994)

Estimation: An alternative approach (CCP)

Idea

Key insight: we can express the value functions in terms of observed choice probabilities. This allows us to obtain the value function without having to solve the agent's optimization problem. This leads to an estimator that is:

- Computationally efficient but **not** econometrically efficient.
- How the mapping from choice probabilities to values is done depends on the application.

Important idea

We don't need to solve for policies whose state variables and outcomes are observed in the data. Therefore, we can simulate play without solving for value functions.

Estimation: An alternative approach (CCP)

Hotz and Miller (1993) and Hotz, Miller, Sanders and Smith (1994)

- Notice that CCP's are unchanged when subtracting a constant from every value:

$$dv(x, a) \equiv v_\theta(x, a) - v_\theta(x, 0),$$

  where $a = 0$ is some reference action.
- Let $Q : \mathbb{R}^{|\mathbf{J}|-1} \to \Delta^{|\mathbf{J}|}$ be the mapping from the deltas in conditional values to CCPs.
- We maintain the assumption that the ϵ are iid.

Hotz and Miller (1993) Proposition 1

Q is invertible.

Estimation: An alternative approach (CCP)

Hotz & Miller inversion with logit errors

Like in Rust, consider $\epsilon \sim$ T1EV, IID. Expression for CCP's:

$$P(a|x;\theta) = \frac{\exp(v_\theta(a,x))}{\sum_{a\in\mathcal{J}} \exp(v_\theta(a,x))}$$

The *HM inversion* follows by taking logs and differencing. Thus, with logit errors:

$$\log P(a|x;\theta) - \log P(0|x;\theta) = v_\theta(x,a) - v_\theta(x,0)$$

Thus, in the logit case

$$Q_i^{-1}(P) = \log P_a - \log P_0$$

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So a possible estimator for θ could be

$$\hat{\theta} = \text{argmin}_\theta || \left[\log(\hat{P}(a=1|x)) - \log(\hat{P}(a=0|x))\right] - [v_\theta(a=1,x) - v_\theta(a=0,x)] ||$$

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The only question is how we compute $v_\theta(a,x)$.

Estimation: An alternative approach (CCP)

Using Inversion for Estimation

The **inversion result** can be used for estimation in a number ways:

1. Forward simulation. *Hotz, et al. (1994)*
2. Finite dependence. *Scott (2016)*.
3. Solve a system of equations to express the vector of values in terms of probabilities and parameters. Pesendorfer and Schmidt-Dengler (2008) for dynamic games.

We will now discuss simulation. For approaches 1-2 we **first** need choice probabilities and transition laws:

$$F(x'|x,a) = \begin{cases} \frac{\sum_{i=1}^{N}\sum_{t=1}^{T-1}\mathbf{1}(x_{i,t+1}<x', x_{it}=x, a_{it}=0)}{\sum_{i=1}^{N}\sum_{t=1}^{T-1}\mathbf{1}(x_{it}=x, a_{it}=0)} & \text{if } a=0 \\ \frac{\sum_{i=1}^{N}\sum_{t=1}^{T-1}\mathbf{1}(x_{i,t+1}<x', a_{it}=1)}{\sum_{i=1}^{N}\sum_{t=1}^{T-1}\mathbf{1}(a_{it}=1)} & \text{if } a=1 \end{cases}$$

$$\hat{P}(a=1|x) \equiv \frac{\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbf{1}(a_{it}=1, x_{it}=x)}{\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbf{1}(x_{it}=x)}, \qquad \hat{P}(a=0|x) \equiv 1-\hat{P}(a=1|x)$$

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Question: Does this work in finite sample for continuous x?

Estimation: An alternative approach (CCP)

Forward Simulation

Two observations:

1. If ϵ is i.i.d. logit, then

$$\mathbb{E}[\epsilon_{a^*}|a = a^*, x] = \gamma - \log(P(a^*|x)),$$

where γ is Euler's constant. For more algebra tricks for discrete choice models have a look at *Discrete Choice Theory of Product Differentiation* by **Anderson, de Palma, and Thisse**.

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Question: What is the intuition for this expectation being decreasing in choice probability?

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$$\mathbb{E}[\epsilon_{a^*}|a = a^*, x] = \gamma - \log(P(a^*|x)),$$

where γ is Euler's constant. For more algebra tricks for discrete choice models have a look at *Discrete Choice Theory of Product Differentiation* by **Anderson, de Palma, and Thisse**.

Question: What is the intuition for this expectation being decreasing in choice probability?

2. With estimates of $\hat{F}(x'|x, a)$, $\hat{P}(a = 1|x)$, $\hat{P}(a = 0|x)$ in hand, we can express the choice specific value function (w/o ϵ) for any a taken this period as follows:

$$v_\theta(a_t, x_t) = u(x_t, a_t; \theta) + \beta \cdot \mathbb{E}\Big[u(x_{t+1}, a_{t+1}; \theta) + \epsilon_{t+1} + \beta \cdot \mathbb{E}[u(x_{t+2}, a_{t+2}; \theta) + \epsilon_{t+2} + \beta^2 \cdot \mathbb{E}[...]]\Big]$$

Estimation: An alternative approach (CCP)

Forward Simulation

Two observations:

1. If ϵ is i.i.d. logit, then

$$\mathbb{E}[\epsilon_{a^*}|a = a^*, x] = \gamma - \log(P(a^*|x)),$$

where γ is Euler's constant. For more algebra tricks for discrete choice models have a look at *Discrete Choice Theory of Product Differentiation* by **Anderson, de Palma, and Thisse**.

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Key idea: Combine these two facts.

Estimation: An alternative approach (CCP)

Inner step: Computing $v_\theta(a,x)$ via forward simulation

For a given θ, simulate forward (for $a = 0, 1$):

$$v_\theta(a,x) \equiv \frac{1}{S}\sum_s \Big[u(a,x;\theta) + \beta \cdot \Big[u(a^s_{t+1}, x^s_{t+1};\theta) + \gamma - \log(\hat{P}(a^s_{t+1}|x^s_{t+1})) + \beta \cdot [u(a^s_{t+2}, x^s_{t+2};\theta) + \gamma - \log(\hat{P}(a^s_{t+2}|x^s_{t+2})) + \beta \cdot ...]\Big]\Big]$$

where

$$x^s_{t+1} \sim \hat{F}(\cdot|x_t, a_t), a^s_{t+1} \sim \hat{P}(\cdot|x^s_{t+1}), x^s_{t+2} \sim \hat{F}(\cdot|x^s_{t+1}, a^s_{t+1}), ...$$

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That is,

- Update state, drawing $x^s_{t+1} \sim F(\cdot \mid x, a)$
- Simulate $t+1$ choice: $a^s_{t+1} = \mathbb{1}\{\text{rand}() < \hat{P}\left(\cdot \mid x^s_{t+1}\right)\}$
- Compute period-$(t+1)$ utility $\beta^1\left(u(a^s_{t+1}, x^s_{t+1};\theta) + \gamma - \log(\hat{P}(a^s_{t+1}|x^s_{t+1}))\right)$
- Update state, drawing $x^s_{t+2} \sim F(\cdot \mid x^s_{t+1}, a^s_{t+1})$; and so on ....

Given simulated choice-specific value functions, predict CCP's:

Estimation: An alternative approach (CCP)

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Given simulated choice-specific value functions, predict CCP's:

$$\tilde{P}(a=1|x;\theta) \equiv \frac{\exp(v_\theta(a=1,x))}{\exp(v_\theta(a=0,x)) + \exp(v_\theta(a=1,x))}$$

Estimation: An alternative approach (CCP)

Full algorithm

- Search over θ to minimize objective

$$|| \left[\log(\hat{P}(a = 1|x)) - \log(\hat{P}(a = 0|x))\right] - [v_\theta(a = 1, x) - v_\theta(a = 0, x)] ||$$

  For each candidate θ,

  - For every (a, x), forward simulate to compute $v_\theta(a, x)$.

Estimation: An alternative approach (CCP)

Full algorithm

The Hotz-Miller estimator:

$$\hat{\theta}_{HM} = \text{argmin}_{\theta} || \left[\log(\hat{P}(a = 1|x)) - \log(\hat{P}(a = 0|x)) \right] - \left[v_{\theta}(a = 1, x) - v_{\theta}(a = 0, x) \right] ||$$

Estimation: An alternative approach (CCP)

Full algorithm

The Hotz-Miller estimator:

$$\hat{\theta}_{HM} = \text{argmin}_{\theta} || \left[\log(\hat{P}(a=1|x)) - \log(\hat{P}(a=0|x)) \right] - \left[v_{\theta}(a=1,x) - v_{\theta}(a=0,x) \right] ||$$

Questions:

- Consider $\hat{P}$, v. Which of these objects do we know and which do we have to compute?
- How do we compute this object?
  - How do we simulate actions?
  - How do we simulate state transitions?
  - Given simulated states and actions, how can we compute payoffs?

Aguirregabiria and Mira (2002)

From Two-Step to K-Step estimators

The CCP approach is sometimes referred to as **two-step** method. **First**, estimate choice probabilities. **Second**, estimate structural parameters.

Note that we could "keep going": Given simulated choice-specific value functions, predict CCP's

$$\tilde{P}(a = 1|x; \theta) \equiv \frac{\exp(v_\theta(a = 1, x))}{\exp(v_\theta(a = 0, x)) + \exp(v_\theta(a = 1, x))}$$

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- CCP estimators reduce computational burden at the expense of econometric efficiency.
- Aguirregabiria and Mira (2002) proposes a K-step estimator to leverage the CCP approach but obtain a more efficient estimator through iteration.
- **Idea:** After forming an initial $\hat{v}(\widehat{P}_0; \theta)$ one can obtain a new set of choice probabilities $\hat{P}_1$ using current guess $\hat{v}$. One can iterate on this and form K-step estimator $\hat{P}_K$.
- The $K = 1$ estimator corresponds to Hotz and Miller and $K = \infty$ to Rust.
- They show that in small samples, each step leads to an increase in efficiency.
- They also show that asymptotically, all K-step estimators are equivalent to MLE.

Estimation: An alternative approach (CCP)

Recap

- **Advantage:** Estimate dynamic parameters without solving dynamic programming problem for each evaluation of θ. The key here is having a consistent estimate of $\hat{P}(a|x)$.
- Leverages following idea:
  - Assume that agents choose the optimal policy in each state
  - We recover "reduced form" consistent estimates of these optimal policies $\hat{P}$ and transitions over state variables $\hat{F}$
  - At θ_0, simulated estimates of choice-specific value functions yield predicted optimal policies $\tilde{P}$ that coincide with observed policies $\hat{P}$.
- When S is discrete, solving for the value function given estimated policies is equivalent to solving a system of linear equations (matrix inversion); see Pesendorfer and Schmidt-Dengler (2008), or use finite dependence, both of which we will discuss later in class.
- CCP approach can be extended to continuous time models, Arcodiacono et al. (2016).

Mathematical Programming with Equilibrium Constraints

Yet another approach for Estimation

Mathematical Programming with Equilibrium Constraints (MPEC)

Su and Judd (2012): *Constrained optimization approaches to estimation of structural models*.

- NFXP not computationally efficient, but CCP not asymptotically efficient.
- In a nested NFXP we require the equilibrium condition to hold for *every* candidate θ.
- **Key idea of MPEC**: Bellman equation can be imposed as a constraint on the optimization.
- As optimization climbs the hill, this constraint can be slack and is only required to hold at the end. $\rightarrow$ many fewer evaluations of Bellman eq.
- Makes use of modern solvers (IPOPT (open source), KNITRO (commercial)).

MPEC

Main Idea

- For NFXP (Rust), likelihood is formulated as $\max_\theta \mathcal{L}(\theta, EV(\theta), \mathbf{X})$, where $EV(\theta)$ is the ex-ante expected value function, and $\mathbf{X}$ is the data.
- However, we know that $EV(\theta)$ is the unique solution to $EV = T(EV, \theta)$
- Use this insight to instead, solve:

$$\max_{EV,\theta} \mathcal{L}(\theta, EV, \mathbf{X}), \text{s.t. } EV = T(EV, \theta)$$

- Same idea can be applied to BLP, using contraction mapping for δ.

Su and Judd Proposition 1

Let $\hat{\theta}$ be the maximum likelihood estimator and let $(\bar{\theta}, \bar{\sigma})$ be a solution of the constrained optimization problem. Define $\hat{\sigma}^\star(\theta) = argmax_{\hat{\sigma}(\theta)} L(\theta, \hat{\sigma}(\theta))$. Then $L(\bar{\theta}, \bar{\sigma}) = L(\hat{\theta}, \hat{\sigma}^\star(\hat{\theta}))$. If the model is identified, then we have $\hat{\theta} = \bar{\theta}$.

Concretely, this means that we can instead solve:

$$\max_{\theta_1, RC, \mathbf{EV}} \mathcal{L}(\theta, RC) = \prod_t P(a_t|x_t, \theta_1)$$

subject to:

$$P(a|x, \theta_1, RC) = \frac{exp(u(x, a; \theta_1, RC) + \beta \int EV_\theta(x')dF(x' \mid x, a))}{\sum_{j=0,1} exp(u(x, j; \theta_1, RC) + \beta \int EV_\theta(x, j)dF(x' \mid x, j))}$$

and

$$EV_\theta(x, a) = T_\theta(EV_\theta)(x, a) = \log\Big(\sum_{a'=0,1} \exp\left(u(x, a; \theta_1, RC) + \beta \int EV_\theta(x', a')dF(x' \mid x, a')\right)\Big)$$

Advantages and Disadvantages:

- No natural way to obtain standard errors. Needs Bootstrap procedure.
- MPEC can be very burdensome on memory. Sparsity is key. In a demand setting sparsity is coming from the assumption that markets are independent.
- Becomes really useful when modern solvers are exploited.
  - Automatic differentiation.
  - Aside: Can automatic differentiation be used in the Rust full solution approach?

Finite Dependence

Arcidiacono and Miller's Lemma

Value Function $\leftrightarrow$ Choice Specific Value Function

Choice Probabilities:

$$P(a|x) = \frac{\exp(v(a,x))}{\sum_j \exp(v(j,x))}$$

Inclusive value of state:

$$EV(x) \equiv \log\left(\sum_j \exp(v(j,x))\right) + \gamma$$

Combining the two:

Arcidiacono and Miller's Lemma

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Combining the two:

$$\begin{aligned} EV(x) &\equiv \log\left(\sum_j \exp(v(j,x))\right) + \gamma \\ &= \log\left(\sum_j \exp(v(j,x))\right) + \log(\exp(v(a',x))) - \log(\exp(v(a',x))) + \gamma \quad \text{for any } a \\ &= v(a,x) - \log(P(a|x)) + \gamma \quad \text{for any } a' \end{aligned}$$

Finite Dependence

An even simpler approach

For all a, x, a',

$$v(a,x) = u(a,x) + \beta \left[\mathbb{E}_{x'} \left[v(a',x') - \log(P(a' \mid x')) \mid a, x \right] + \gamma \right]$$

Finite Dependence

An even simpler approach

For all a, x, a',

$$v(a,x) = u(a,x) + \beta \left[\mathbb{E}_{x'} \left[v(a',x') - \log(P(a' \mid x')) \mid a, x\right] + \gamma\right]$$

In the binary case: For all x, a',

$$\begin{aligned} v(1,x) - v(0,x) = u(1,x) - u(0,x) + \\ \beta \mathbb{E}_{x'} \left[v(a',x') - \log(P(a' \mid x')) \mid a = 1, x\right] \\ - \beta \mathbb{E}_{x'} \left[v(a',x') - \log(P(a' \mid x')) \mid a = 0, x\right] \end{aligned}$$

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Now suppose that action $a' = 1$ resets the state (like an engine replacement does for Harold Zuercher).

Finite Dependence

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Now suppose that action $a' = 1$ resets the state (like an engine replacement does for Harold Zuercher). Then,

$$\begin{aligned} \mathbb{E}\left[v(1,x') \mid a = 1, x \right] - \mathbb{E}\left[v(1,x') \mid a = 0, x \right] &= \mathbb{E}\left[u(1,x') \mid a = 1, x \right] - \mathbb{E}\left[u(1,x') \mid a = 0, x \right] \\ &= 0 \text{ (in Harold Zuercher's case, at least)} \end{aligned}$$

Finite Dependence

An even simpler approach

Thus, for Harold Zuercher, for all x,

$$\begin{aligned} v(1,x) - v(0,x) = u(1,x) - u(0,x) \\ + \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 0, x \right] \\ - \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 1, x \right] \end{aligned}$$

So difference is choice-specific value functions is the sum of

1. The difference in current payoffs.
2. The (discounted) (negative) effect of a replacement today on next period's log probability of replacement.

Finite Dependence

An even simpler approach

Stepping back:

We are interested in parameters θ governing current payoffs $u(x, a; \theta)$. The Hotz-Miller inversion does not directly get us to an estimate of θ. We went through the steps to get us there:

1. Express $v(\cdot; \theta)$ in terms of observed choice probabilities $P(\cdot)$ and known current payoffs $u(\cdot, \cdot; \theta)$
2. Construct objective function from implied choice probabilities

Finite Dependence

An even simpler approach

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1. Express $v(\cdot; \theta)$ in terms of observed choice probabilities $P(\cdot)$ and known current payoffs $u(\cdot, \cdot; \theta)$
2. Construct objective function from implied choice probabilities

But in the case of **finite dependence** (Arcidiacono & Miller '11), things are even simpler:

- There are sequences of actions which lead to the same state
- After getting to that state, the continuation values are the same
- Differences in the frequencies of those action sequences are related to differences in current-period payoffs
- And, with finite dependence, we can actually deal with time-varying (aggregate) unobservables.

Finite Dependence

Renewal Action

Suppose $a = 1$ takes you to the same distribution of states, **regardless of the current state**. In Rust, replacing the engine is a renewal action:

- $x_t = \text{mileage}_t = 0$ if $a_t =$ replace, regardless of previous state $x_{t-1} = \text{mileage}_{t-1}$

More generally, we require that $\forall x_t, x_t'$:

$$F(x_{t+1} \mid x_t, a_t = 1) = F(x_{t+1} \mid x_t', a_t = 1)$$

$\rightarrow$ $a_t = 1$ yields the same continuation value.

DDC Estimation via OLS
a.k.a.
“Euler equation approach”
or
“Euler CCP approach”

Finite dependence $\rightarrow$ OLS

Thus, for Harold Zuercher, for all x,

$$\begin{aligned} v(1,x) - v(0,x) = u(1,x) - u(0,x) & \\ + \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 0, x \right] & \\ - \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 1, x \right] & \end{aligned}$$

And choice probabilities are

$$p(x) \equiv \Pr(a = 1 \mid x) = \frac{\exp(v(1,x) - v(0,x))}{1 + \exp(v(1,x) - v(0,x))}$$

Finite dependence → OLS

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$$\begin{aligned} v(1,x) - v(0,x) = u(1,x) - u(0,x) & \\ + \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 0, x \right] & \\ - \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 1, x \right] & \end{aligned}$$

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Thus,

$$\begin{aligned} \log \frac{p(x)}{1 - p(x)} &= v(1,x) - v(0,x) \\ &= u(1,x) - u(0,x) \\ &\quad + \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 0, x \right] \\ &\quad - \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 1, x \right] \end{aligned}$$

Scott (2016)
Overview

Applied question

- What are the effects of biofuels policy?
- US biofuels mandate: $\approx$ 10% of gasoline must come from biofuels.
- Supply elasticities based on static models might be downwards-biased.

Estimate model using **linear regression**.

- Relies on Hotz-Miller (1993) inversion.
- Exploits finite dependence, Altug and Miller (1998).
- Can be seen as an **"Euler equation"** approach for DDC models.

Scott (2016)
Method

Key insight:

- Realizations of agents future utility are noisy measures of agent's expectation.

Advantages of approach:

- No need to specify how all state variables evolve over time.

Can be combined with **quasi-experimental** variation:

- Diamond, McQuade, Qian (2018): combine with diff-in-diff IV to study rent control.
- No need to explicitly model expectations.

Scott (2016)
Model + Assumptions

Binary crop choice model
A landowner's choice set: $\mathcal{A} = \{crops, other\}$. If field i is in state x at time t, then the expected profits to land use j are:

$$\pi\left(a, x, \omega_t, \nu_{it}\right) = \alpha_{0,a,x} + \alpha_R \cdot R_a\left(\omega_t\right) + \xi_{a,x}\left(\omega_t\right) + \nu_{iat}$$

Notation: i: field,a: land use, x: field state, ω: market state, R: expected returns, observable to econometrician, ξ: unobservable shock to returns, ν: idiosyncratic field-level shock, α: parameters to be estimated,
Assumptions:

1. Small fields, no externalities: the distribution of the market state ω_{t+1} conditional on ω_t is not affected by changing the land use in any single field.
2. Logit errors: The idiosyncratic error term ν_{iat} has a type 1 extreme value distribution, independently and identically distributed across i, a, and t.
3. Landowners have rational expectations.

Scott (2016)

Dynamics: Field-Level

Landowners maximize expected discounted profits. Field states evolve according to a simple deterministic process:

$$x_{i,t+1} = x^{+}\left(a_{it}, x_{it}\right) = \begin{cases} 0 & \text{if } a_{it} = \textit{crops} \\ \min\{x_{it} + 1, \bar{x}\} & \text{if } a_{it} = \textit{other} \end{cases}$$

Important details:

- Like in bus-engine replacement, there is a renewal action.
- Unlike bus-engine replacement, transition process for x here is *deterministic* and state is *discrete*.
  - → However, there's still a stochastic transition process for the *aggregate*, *exogenous* state variable ω.
- No explicit assumptions on the evolution of R and ξ
- Estimating the process governing the evolution of the unobservable supply shock ξ is especially difficult.

Scott (2016)

Regression equation construction

Recall that for Rust we derived

$$\begin{aligned}\log \frac{p(x)}{1-p(x)} &= v(1,x) - v(0,x) \\ &= u(1,x) - u(0,x) + \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 0, x\right] \\ &\qquad - \beta \mathbb{E}_{x'} \left[\log(P(a' \mid x')) \mid a = 1, x\right]\end{aligned}$$

Mapping this to Scott's model, we have

$$\begin{aligned}\log \frac{p_{\text{crops}}(x, \omega_t)}{1-p_{\text{crops}}(x, \omega_t)} = \alpha_{0,1,x} - \alpha_{0,0,x} &+ \alpha_R \left(R_1(\omega_t) - R_0(\omega_t)\right) \\ &+ \left(\xi_{1,x}(\omega_t) - \xi_{0,x}(\omega_t)\right) \\ &+ \beta \mathbb{E}_{\omega_{t+1}} \left[\log p_{\text{crops}}(\min\{x_{it}+1, \bar{x}\}, \omega_{t+1})\right] \\ &\quad - \beta \mathbb{E}_{\omega_{t+1}} \left[\log p_{\text{crops}}(0, \omega_{t+1})\right]\end{aligned}$$

Scott (2016)

Regression equation construction

$$\begin{aligned}\log \frac{p_{\text{crops}}(x, \omega_t)}{1 - p_{\text{crops}}(x, \omega_t)} = \alpha_{0,1,x} - \alpha_{0,0,x} + \alpha_R \left(R_1(\omega_t) - R_0(\omega_t)\right) \\ + \left(\xi_{1,x}\left(\omega_t\right) - \xi_{0,x}\left(\omega_t\right)\right) \\ + \beta \mathbb{E}_{\omega_{t+1}}\left[\log p_{\text{crops}}(\min\left\{x_{it} + 1, \bar{x}\right\}, \omega_{t+1})\right] \\ - \beta \mathbb{E}_{\omega_{t+1}}\left[\log p_{\text{crops}}(0, \omega_{t+1})\right]\end{aligned}$$

Note: We don't know expectation of these log probabilities, but we do know their *realizations*. And realizations are a noisy measure of expectations:

$$\begin{aligned}\mathbb{E}_{\omega_{t+1}}\left[\log p_{\text{crops}}(\min\left\{x_{it} + 1, \bar{x}\right\}, \omega_{t+1}) - \log p_{\text{crops}}(0, \omega_{t+1})\right] = \\ \log p_{\text{crops}}(\min\left\{x_{it} + 1, \bar{x}\right\}, t + 1) - \log p_{\text{crops}}(0, t + 1) + \frac{1}{\beta}\Delta\epsilon_t\end{aligned}$$

where $\Delta\epsilon$ is the *expectational error*, which is independent of x, ω_t.

Scott (2016)

Regression equation construction

$$\underbrace{\log \frac{p_{\text{crops}}(x,t)}{1-p_{\text{crops}}(x,t)}}_{Y_{x,t}} = \Delta\alpha_0(x) + \alpha_R \Delta R_t + \Delta\xi_t(x) + \beta \log p_{\text{crops}}(\min\{x_{it}+1, \bar{x}\}, t+1) - \beta \log p_{\text{crops}}(0, t+1) + \Delta\epsilon_t$$

Rearranging gives

$$\log \frac{p_{\text{crops}}(x,t)}{1-p_{\text{crops}}(x,t)} + \beta \log \frac{p_{\text{crops}}(0,t+1)}{p_{\text{crops}}(\min\{x_{it}+1, \bar{x}\}, t+1)} = \Delta\alpha_0(x) + \alpha_R \Delta R_t + \Delta\xi_t(x) + \Delta\epsilon_t$$

So regressing Y on ΔR and a set of xt fixed effects enables one to estimate α_R.

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So regressing Y on ΔR and a set of xt fixed effects enables one to estimate α_R.

Practical issues: What if we have only a few observations for some x, t? What if we only ever observe $a = 0$ or we only ever observe $a = 1$ for some x, t?

Identification in DDC

Digression: Nonparametric Identification

Definition

Question: What do we mean by identification?

Digression: Nonparametric Identification

Definition

Question: What do we mean by identification?

- Let $m^\star$ be the true vector of functions and distributions.
- Let $P(m)$ denote the joint distribution of observable variables under the assumption that the data is generated under m

Definition of Identification

Definition 1: $m^\star$ is identified in M iff $\forall m \in M,\ m \neq m^\star,\ P(m) \neq P(m^\star)$

Definition 2: $m^\star$ is not identified in M iff $\exists m \in M,\ m \neq m^\star,\ P(m) = P(m^\star)$

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- If M is a subset of a finite dimensional space, we say that the model M is **parametric**.
- If M is not a subset of a finite dimensional space, we say that M is:
  - **Semi-parametric**, if some of the functions, distributions lie inside a finite dimensional space.
  - **Nonparametric**, if none of the functions, distributions lie inside a finite dimensional space.
- Notice, that we are not worried about finite sample variation. Identification asks, if we had infinite data, can we recover the objects of interest.

Refresher: Nonparametric Identification

Some Comments

- For many canonical models (such as BLP, Rust, and many types of auction setups) non-parametric identification has been shown.
- If you want to be creative and come up with your own model some people will ask how it is non-parametrically identified.

Refresher: Nonparametric Identification

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Two popular ways to prove identification.

(1) Take m' and $m^\star$, and proceed as in definition, i.e., prove that $P(m') \neq P(m^\star)$ if $m' \neq m^\star$.

(2) Express m as a functional of $P(m)$ for all $m \in M$, as $m = T(P(m))$ for some T.

The latter "constructive approach" is preferred since it often provides a natural estimator for free.

Refresher: Nonparametric Identification

Some comments

Identification and Modeling: When we make an economic model that we wish to take to the data, we have to think of two things:

(1) How well does the model capture important economic behavior?

(2) Can we identify and estimate the model using the data we have?

(1) tends to push your model to become more and more complicated. (2) pushes your model to become simpler. Your model building exercise will be a constrained optimization problem: Maximize (1) subject to obtaining (2). Knowing the limits of identification allows you to max out on (1), which is a good thing.

Nonparametric Identification

Discrete Choice, $\epsilon \sim T1EV$

Consider the following **Discrete Choice** model:

$$U_j = u_j(X) + \epsilon_j$$
$$Y = arg \max_{j \in J} \{U_j\}$$

Assume: $\epsilon \perp X$, and that we observe (Y, X_j), whereas ϵ_j is unobservable. For now, look at case where ϵ_j's are independent across j. Q: can we identify $(u(.), F_{\epsilon_1,...,\epsilon_J})$.? What we can conclude right away:

- We cannot identify the scale of u_j separately from F_ϵ. Consider two models (u_j, F_ϵ) and $(C \cdot u_j, F_{C\epsilon})$, where $F_{C\epsilon}(\tau) = F_\epsilon(\tau/C)$.
- We cannot identify u_j and ϵ up to a constant (can have $u_j = u_j - \kappa$ and $\varepsilon = \varepsilon + \kappa$ so that κ cancels out). Consider (u_j, F_ε) and $(u_j - \kappa, F_{\varepsilon+\kappa})$ where $F_{\varepsilon+\kappa}(\tau) = F_\varepsilon(\tau - \kappa)$.
- We cannot identify u_j up to an additive function of X. Consider (u_j, F_ϵ) and $(u_j + g(X), F_\epsilon)$, where $g(.)$ does not depend on j. We can set $u_J = 0$ without loss of generality.

Let's consider the case where $\mathbb{E}[\epsilon_j] = 0$, $u_0 = 0$, $\mathbb{V}[\epsilon_j] = \sigma = 1$, $\epsilon \sim T1EV$. **Claim:** u_j is non-parametrically identified.

Nonparametric Identification

Discrete Choice, $\epsilon \sim T1EV$

Proof:

$$\log(P(Y = j|X)) = \log\left(\frac{exp(U_j)}{\sum_{k\in J}\exp(U_k)}\right) = U_j - \log\left(\sum_k \exp(U_k)\right)$$

And therefore:

$$\log(P(Y = j|X)) - \log(P(Y = 0|X)) = U_j - U_0 = u_j(X) - u_0(X) = u_j(X)$$

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Estimation:
If X is discrete:

$$\log(\widehat{P_j(x)}) - \log(\widehat{P_J(x)}) = \log\Big(\frac{\sum 1\{Y=j \wedge X=x\}}{\sum 1\{X=x\}}\Big) - \log\Big(\frac{\sum 1\{Y=j \wedge X=x\}}{\sum 1\{X=x\}}\Big) \xrightarrow{P} u_j(x),$$

else use a Kernel estimator or sieves.

Nonparametric Identification

Discrete Choice, $\epsilon \sim T1EV$

- Hotz & Miller (1993), as well as Magnac & Thesmar (2002) extend the above result for static models to dynamic models.
- We already knew this, since we have expressed value functions in terms of choice probabilities for estimation (**H+M** inversion).
- More specifically: we can specify a vector of utilities for the reference action π, a distribution for the idiosyncratic shocks G, and a discount factor β, and we will be able to find a model rationalizing the observed CCPs that features (π, β, G) (**Magnac and Thesmar (2002)**).
- This implies that the discount factor is not identified.

Identification, Dynamic Single Agent Models

Intuitive explanation

H+M: $\log(p_j(x)) - \log(p_J(x)) = v_i(x) - v_J(x) \;\; \forall \;\; j \in \mathcal{J} \backslash J, x \in X$. This summarizes all the predictions the model makes about the data. Need β to decompose $v_j(x) - v_J(x)$ into a current utility and a continuation value component.

- $(J-1) \times |X|$ equations in **H+M**.
- $(J-1) \times |X| + 1$ unknown primitives, $\Delta \mathbf{v}$ and β.

Solution:

- Tests for forward-looking behavior exploit scenarios in which some variables which affect future utility are known in period t: consumers are deemed forward-looking if their period t decisions depends on these variables. (Example: Chevalier and Goolsbee (2005))
- Use exclusion restriction for set identification, shift continuation value component, holding current period utility fixed (Abbring and Oeystein (2018)).
- Typically, literature treats β as known.

Persistent Unobserved Heterogeneity

Arcidiacono and Miller (2011)

Persistent unobserved heterogeneity

So far: Unobserved heterogeneity (in form of action specific ϵ) has convenient form.

- Temporary, IID.

Distinguish two cases:

- Unobserved heterogeneity has some persistence and might change over time. If you are interested in a serious econometric account of such "Hidden Rust" models, see Connault (2016).
- Unobserved heterogeneity fully persistent. Formal identification papers: Hall and Zhou (2003), Kasahara and Shimotsu (2009).

Arcodiacono and Miller (2011) provide model for fully persistent heterogeneity.

- Examples in Harold Zurcher setting:
  - Bus brand unobserved to econometrician.
  - Harold Zurcher's son makes decisions for some (unknown) subset of buses.
- **CCP approach:** if there is an unobserved state s, $p(x, a)$ no longer an unbiased estimator.

Arcodiacono and Miller (2011)

The problem

Setup: Start again from Rust (1987), add unobserved (to the econometrician) persistent state $s \in \{1, 2, \ldots, S\}$.

- Persistent unobserved heterogeneity at the bus level. Denote π_s the population frequency of type s.
- Each s has its own corresponding payoff parameters θ_s.

Notation: i indexes buses, t indexes time, $a_{it} \in \{0, 1\}$ is action, $x_{it} \in \mathcal{X}$ is state, and $p(x; \theta_s) = \Pr(a = 1 \mid x; \theta_s)$.

Question: How do we formulate the log-likelihood?

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$$\{\hat{\theta}, \hat{\pi}\} = \arg\max_{\theta, \pi} \sum_i \log \left(\sum_{s=1}^{S} \pi_s \prod_t \mathcal{L}(a_{it} | x_{it}, \theta_s) \right)$$

Questions:

- Do you foresee any issues with solving this optimization problem?

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Questions:

- Do you foresee any issues with solving this optimization problem?
- Does this likelihood formulation look familiar?

Arcodiacono and Miller (2011)

EM Algorithm

$$\{\hat{\theta}, \hat{\pi}\} = \arg\max_{\theta,\pi} \sum_i \log\Big(\sum_{s=1}^{S} \pi_s \prod_t \mathcal{L}(a_{it}|x_{it}, \theta_s)\Big)$$

EM Algorithm: instead of directly optimizing the above, iterate on the estimation of $\hat{\theta}$ and $\hat{\pi}$. Assume we have some initial guess of $\{\pi_s\}$ and $\{\theta_s\}$.

This implies choice probabilities, and then via Bayesian updating, we can compute

$$q_{is} = P(s_i = s|a_i, x_i; \theta, \pi) = \frac{\pi_s \cdot \prod_t \mathcal{L}(a_{it}|x_{it}, \theta_s)}{\sum_{k=1}^{S} \pi_k \cdot \prod_t \mathcal{L}(a_{it}|x_{it}, \theta_k)}$$

and from that:

$$\hat{\pi}_s = \frac{1}{I} \sum_{i=1}^{I} \hat{q}_{is}$$

Arcodiacono and Miller (2011)

EM Algorithm

Algorithm, Iterate on (1) to (3) until convergence.:

(0) Initialize with some weights and parameters.

Iterate: For the mth step:

(1) $\theta^{(m+1)} = \arg\max_\theta \sum_i \sum_s \sum_t q_{is}^{(m+1)} \log\left(\mathcal{L}(a_{it}|x_{it}, \theta_s^{(m)})\right)$

(2) $q_{is}^{(m+1)} = \frac{\pi_s^{(m)} \prod_t \mathcal{L}\left(a_{it}|x_{it}, \theta_s^{(m)}\right)}{\sum_{k=1}^{S} \pi_k^{(m)} \prod_t \mathcal{L}\left(a_{it}|x_{it}, \theta_k^{(m)}\right)}$

(3) $\pi_s^{(m+1)} = \frac{1}{I} \sum_i q_{is}^{(m+1)}$

Note on EM algorithm:

- Has the property that each iteration increases the likelihood.
- But no guarantee of convergence, i.e. might get stuck in local max. **Question**: How do we deal with this?
- Convergence can be very slow.
- Bonhomme, et al. (2017) "Discretizing Unobserved Heterogeneity" as an alternative to EM algorithm.

Autocorrelated Unobservables

Some other relevant work

Related work to take care of endogeneity issues caused by **autocorrelated** unobserved state variables:

Berry and Compiani (2017): *An Instrumental Variable Approach to Dynamic Models*

- Generalized instrumental variable approach (Chesher and Rosen (2015)).
- Set identification, estimator is discrete grid search.

Kalouptsidi et Al. (2017): *An Instrumental Variable Approach to Dynamic Models*

- Linear IV approach builds on Scott's Euler equations insight.